

# Riemannian and Complex Geometry

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Submission deadline: 24 January 2025, 10:00 am (GMT-4)

— Examen Final —

## Rules for the final exam:

- Submission deadline: **24 January 2026, 10:00 am (GMT-4)**.
- Late submission of the exam may result in point deductions or a complete annulment of the exam grade.
- Complete **only three (3)** of the following five (5) exercises.
- Submit your solutions in an organized manner; solutions that cannot be understood will not be graded.

### 1.1 Problem 1

Let  $(M, g)$  be a Riemannian manifold. Prove that the scalar curvature  $\text{scal}(p)$  at  $p \in M$  is given by

$$\text{scal}(p) = \frac{1}{\omega_{n-1}} \int_{\mathbb{S}^{n-1}} \text{Ric}_p(x, x) dS$$

where  $\omega_{n-1}$  is the area of the sphere  $\mathbb{S}^{n-1}$  in  $T_p M$  and  $dS$  is the area elements on  $\mathbb{S}^{n-1}$ .

### 1.2 Problem 2

By analogy with the formula  $\text{div} X = \text{tr}(\nabla X)$ , we can define a divergence operator on tensor fields of any rank on a Riemannian manifold. If  $F$  is any smooth  $k$ -tensor field (covariant, contravariant, or mixed), we define the divergence of  $F$  by

$$\text{div} F = \text{tr}_g(\nabla F),$$

where the trace is taken on the last two indices of the  $(k+1)$ -tensor field  $\nabla F$ . Extend the integration by parts formula as follows: if  $F$  is a smooth covariant  $k$ -tensor field and  $G$  is a smooth covariant  $(k+1)$ -tensor field on a compact smooth Riemannian manifold  $(M, g)$  with boundary, then

$$\int_M \langle \nabla F, G \rangle dV_g = \int_{\partial M} \langle F \otimes N^b, G \rangle dV_{\hat{g}} - \int_M \langle F, \operatorname{div} G \rangle dV_g,$$

where  $N$  is the outward-pointing unit normal vector field along  $\partial M$  and  $\hat{g}$  is the induced metric on  $\partial M$ .

### 1.3 Problem 3

Let  $(M, g)$  be a compact Riemannian manifold of dimension  $n$ . Suppose

$$\operatorname{Ric}(g) \geq \lambda g > 0$$

Show that the first nonzero eigenvalue  $\lambda_1$  of the Laplacian operator  $\Delta = dd^* + d^*d$  satisfies

$$\lambda_1 \geq \frac{n}{n-1} \lambda.$$

Moreover, show that the equality holds if and only if  $\operatorname{Hess} f = -\frac{\Delta f}{n} g$ .

### 1.4 Problem 4

Let  $(X, \omega)$  be a Hermitian manifold. There is a natural map

$$\begin{aligned} f: H_{dR}^1(X, \mathbb{R}) &\rightarrow H^{0,1}(X) \\ [\eta] &\mapsto [\eta^{0,1}]. \end{aligned}$$

Prove  $f$  is an injective  $\mathbb{R}$ -linear map and thus conclude  $\dim_{\mathbb{R}} H_{dR}^1(X, \mathbb{R}) \leq \dim_{\mathbb{R}} H^{0,1}(X)$ .

### 1.5 Problem 5

Let  $(X, \omega)$  be a compact Kähler manifold. If  $(L, h)$  is a semi-positive line bundle, that is, its Chern curvature is semi-positive definite, and suppose the rank of its Chern curvature is  $\geq k$ , then for any  $p + q \geq 2n - k + 1$ , we have

$$H^{p,q}(X, L) = 0.$$

In particular, if  $(L, h)$  is positive, then  $k = n$ , and it reduces to Kodaira vanishing theorem.